

**Q1. Harshad number****ANSWER**

```
n = int(input("Enter a number: "))
s = sum(int(i) for i in str(n))
if n % s == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")
print("\n Harshad Numbers between 1 and 500:")
for i in range(1, 501):
    if i % sum(int(j) for j in str(i)) == 0:
        print(i, end = " ")
```

**Q2. GCD****CODE**

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = a
    a = b
    b = temp % b
print("GCD is", a)
```

**INPUT**

24  
18

**OUTPUT**

Enter first: Enter second: GCD is 6

**Q3. sum to a target value****ANSWER**

```
def find_pairs(1st, target):
    pairs = []
    for i in range(len(1st)):
        for j in range(i + 1, len(1st)):
            if 1st[i] + 1st[j] == target:
                pairs.append((1st[i], 1st[j]))
    return pairs
data = list(map(int, input("Enter numbers: ").split()))
t = int(input("Enter target: "))
print("Pairs:", find_pairs(data, t))
```

**Q4. Bitwise Operators****ANSWER**

```
a = 0b1101
b = 0o25
c = 0x1a

result = (a & b) | c

print(f"a={a}, b={b}, c={c}, result={result} ")
print(bin(result), oct(result), hex( result))
print(result >> 2, result << 1, ~result)
```

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**Q5. binary search****CODE**

```
def rotate_binary_search(arr, target):  
    low = 0  
    high = len(arr) - 1  
  
    while low ≤ high:  
        mid = (low + high) // 2  
  
        if arr[mid] == target:  
            return mid  
  
        if arr[low] ≤ arr[mid]:  
            if target ≥ arr[low] and target < arr[mid]:  
                high = mid - 1  
            else:  
                low = mid + 1  
        else:  
            if target > arr[mid] and target ≤ arr[high]:  
                low = mid + 1  
            else:  
                high = mid - 1  
  
    return -1  
  
data = [4, 5, 6, 7, 0, 1, 2]  
t = int(input("Target: "))  
  
result = rotate_binary_search(data, t)  
  
if result ≠ -1:  
    print("Found at index:", result)  
else:  
    print("Target not found")
```

**OUTPUT**

Target: Found at index: 4